COMPUTER NETWORKS

CSE 321

IoT

The Internet of Things (IoT) is a network of physical objects embedded with sensors, software, and communication technologies, enabling them to connect to the Internet and exchange data automatically without direct human involvement.

Motivation

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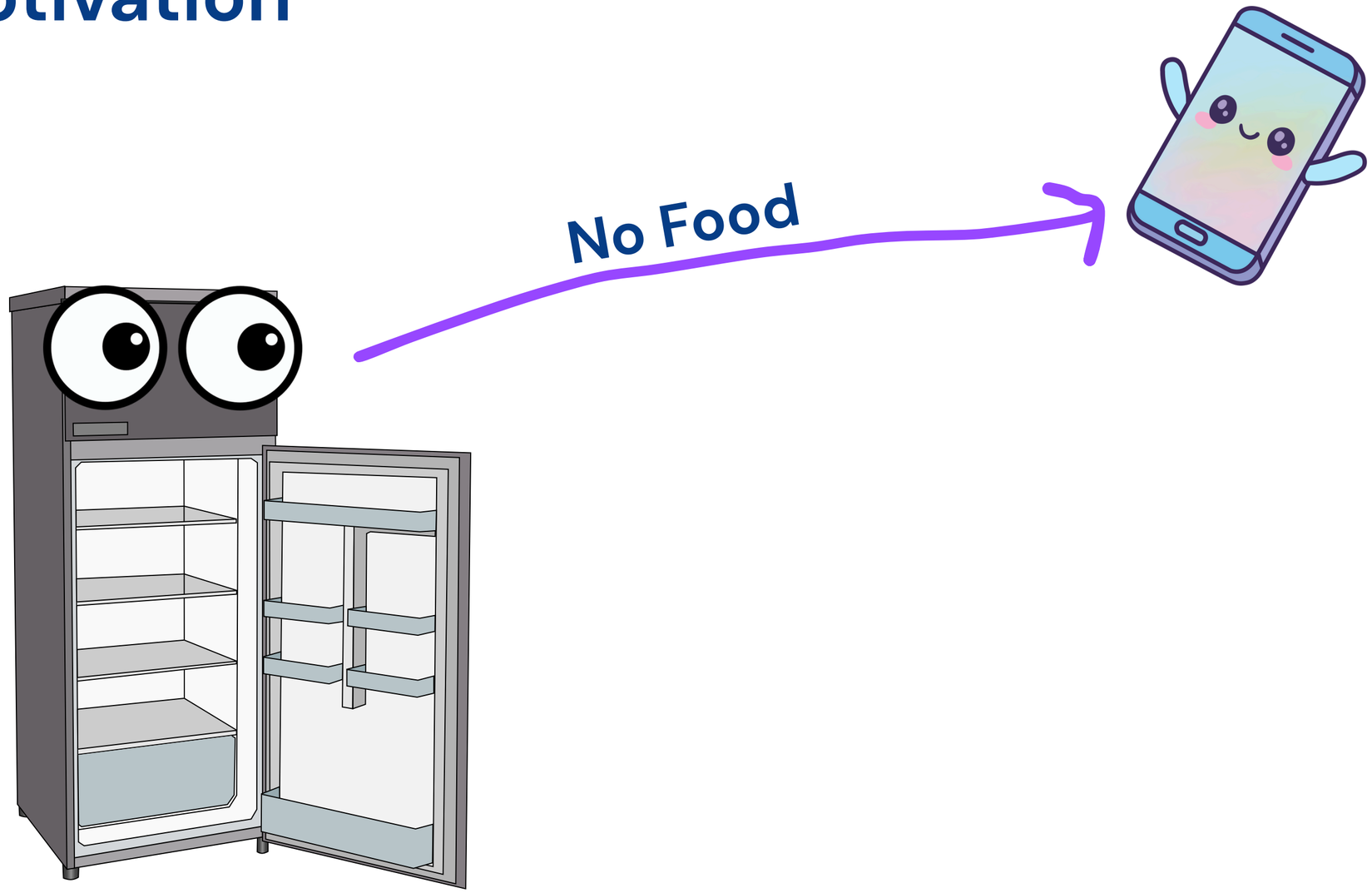

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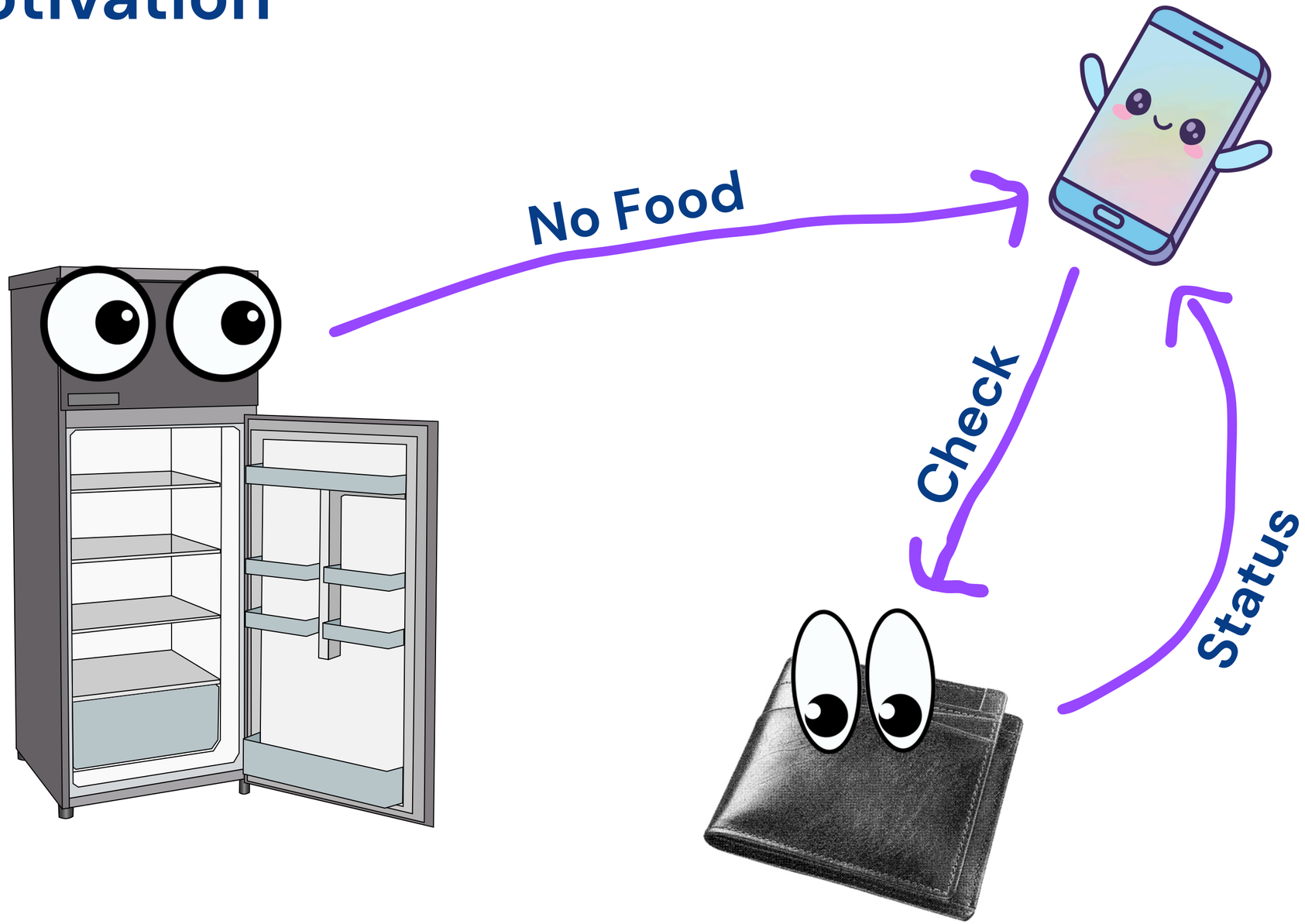

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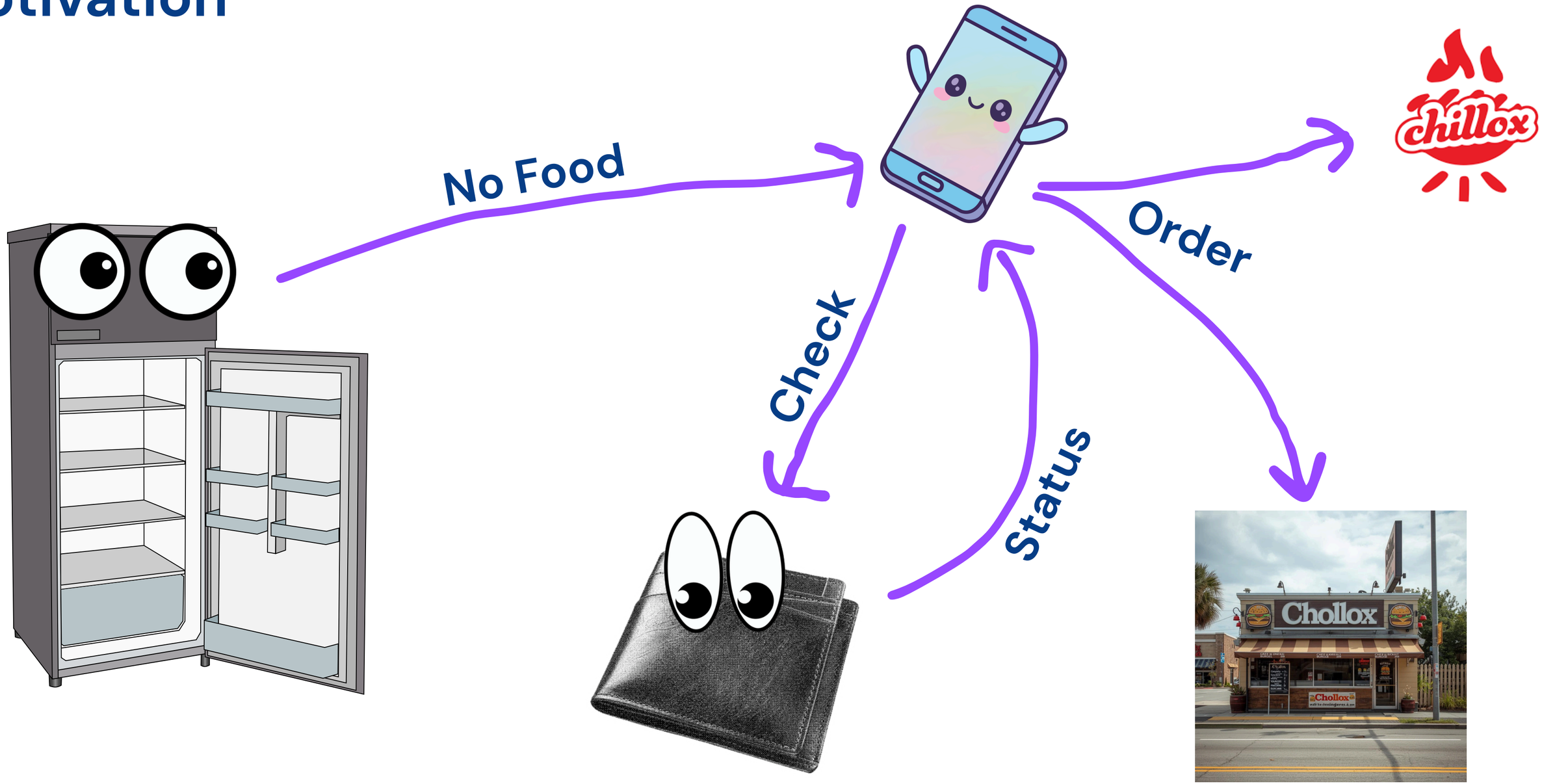


Motivation

- Operational Efficiency & Cost Savings
- Data-Driven Decision Making
- Enhanced Convenience & Quality of Life
- Real-time monitoring

Components of IoT System

- **IoT Devices –** Physical objects such as appliances, machines, vehicles, or wearables that sense data from the environment or perform actions.

Components of IoT System

- IoT Devices
- Sensors and Actuators – Sensors collect environmental data like temperature, motion, or light, while actuators convert digital signals into physical actions.

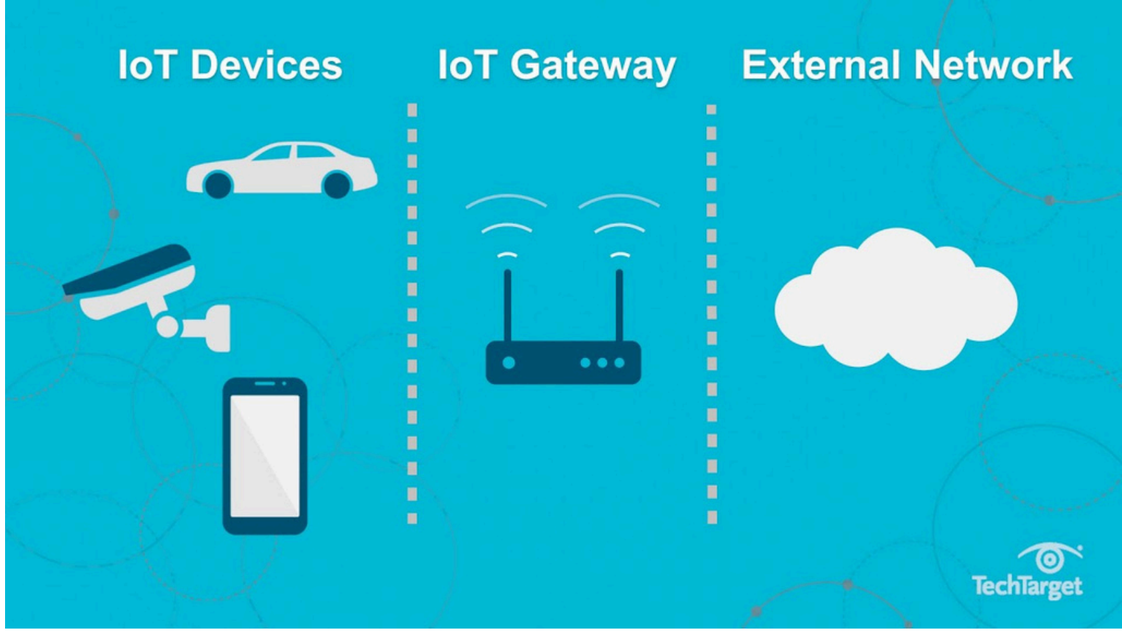
Components of IoT System

- IoT Devices
- Sensors and Actuators
- Embedded Processing Units – Microcontrollers or microprocessors that process sensor data, execute control logic, and manage device operations.

Components of IoT System

- IoT Devices
- Sensors and Actuators
- Embedded Processing Units
- Connectivity Modules – Enable devices to communicate over networks using technologies such as Wi-Fi, Bluetooth, cellular networks, or LPWAN.

Components of IoT System



- IoT Devices
- Sensors and Actuators
- Embedded Processing Units
- Connectivity Modules
- IoT Gateways – Aggregate data from multiple devices, perform protocol translation, and securely transmit data to cloud or edge platforms.

Components of IoT System

- IoT Devices
- Sensors and Actuators
- Embedded Processing Units
- Connectivity Modules
- IoT Gateways
- Cloud / Edge Computing – Cloud platforms store and analyze large IoT data, while edge computing processes data near the source to reduce latency.

Components of IoT System

- IoT Devices
- Sensors and Actuators
- Embedded Processing Units
- Connectivity Modules
- IoT Gateways
- Cloud / Edge Computing
- User Interface and Applications – Dashboards, mobile apps, or web interfaces that allow users to monitor data, control devices, and receive alerts.

IoT Architecture

Depending on functionality, workflow or architecture varies.

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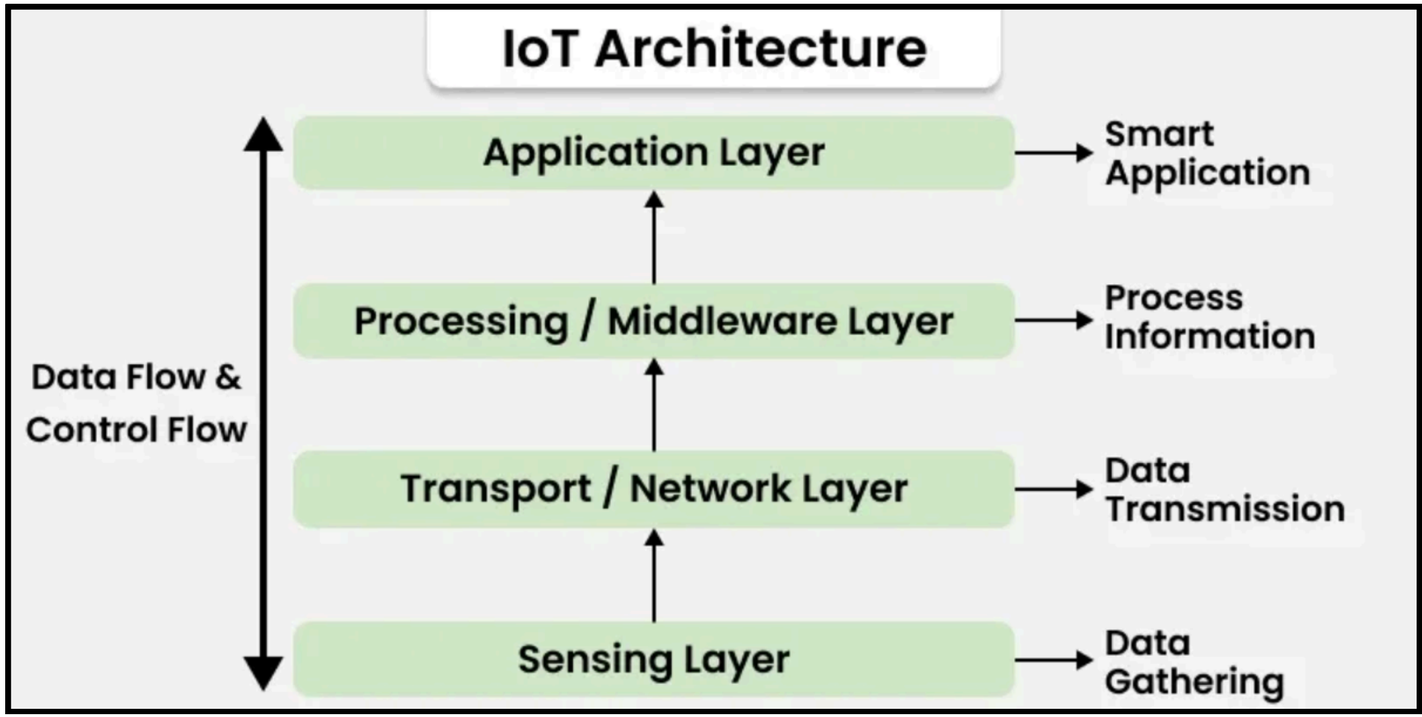
Basic architecture involves 4 layers:

- Sensing Layer
- Network Layer
- Data Processing Layer
- Application Layer



Perception / Sensing Layer

- Collects real-world data such as temperature, humidity, motion, light, or pressure using sensors.
- Uses actuators to perform physical actions like switching devices on/off or adjusting system parameters.

Transport / Network Layer

- Transfers sensed data from IoT devices to gateways, edge nodes, or cloud platforms.
- Maintains reliable communication using wired and wireless technologies like Wi-Fi, Bluetooth, Zigbee, and cellular networks.
- Supports communication protocols to ensure efficient and secure data exchange between devices and systems.

Processing / Middleware Layer

- Filters, analyzes, and stores incoming data for further use and decision-making.
- Manages connected devices, handles interoperability, and supports scalability across large IoT deployments.
- Ensures data integrity, authentication, and secure access to IoT services.

Application Layer

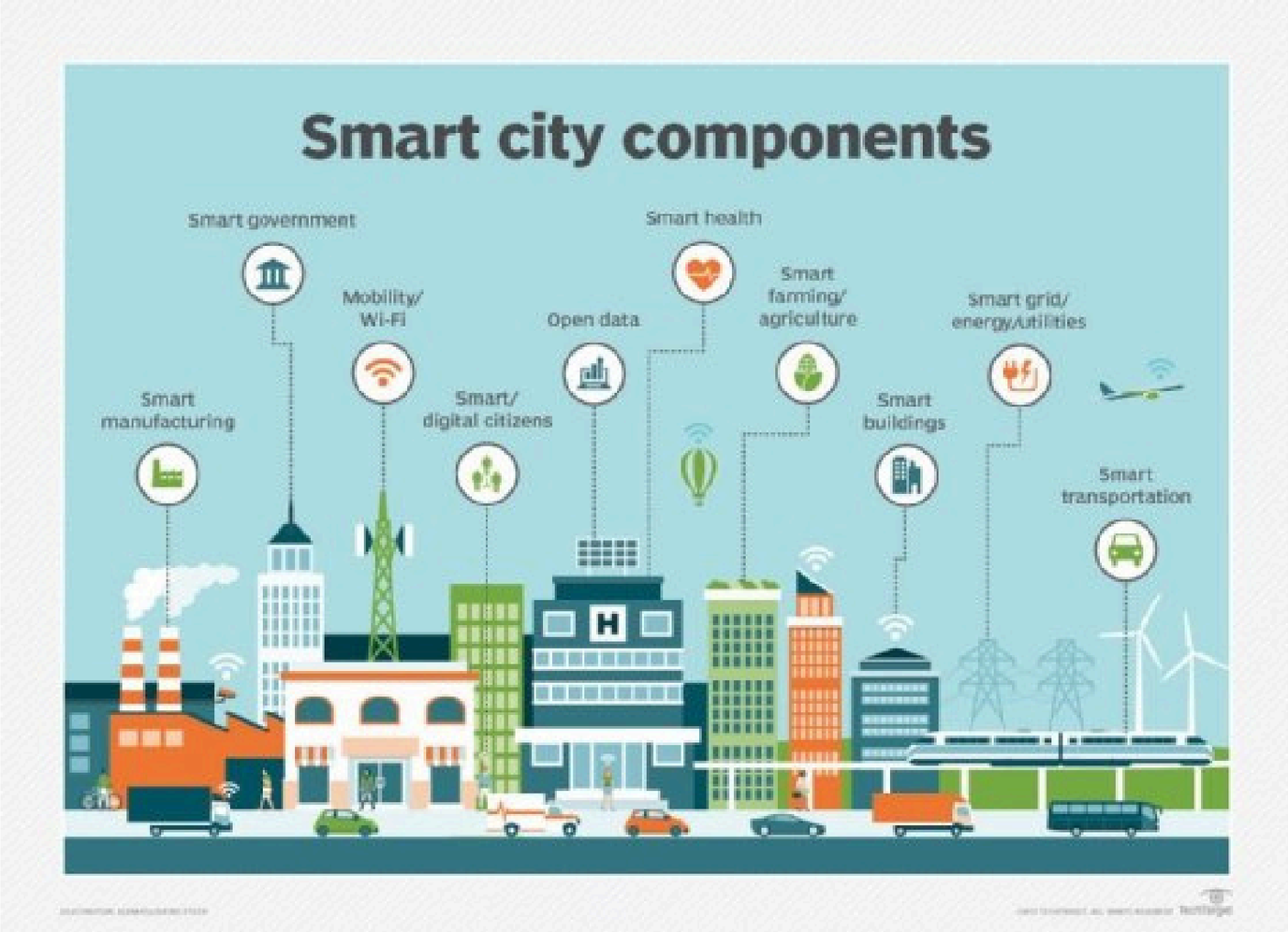
- Provides dashboards, mobile apps, or web interfaces for monitoring and controlling IoT systems.
- Presents processed data in a meaningful form to support analysis and decision-making.

Applications

Smart Homes: Control lighting, heating, appliances, and security systems remotely for comfort, energy savings, and safety.

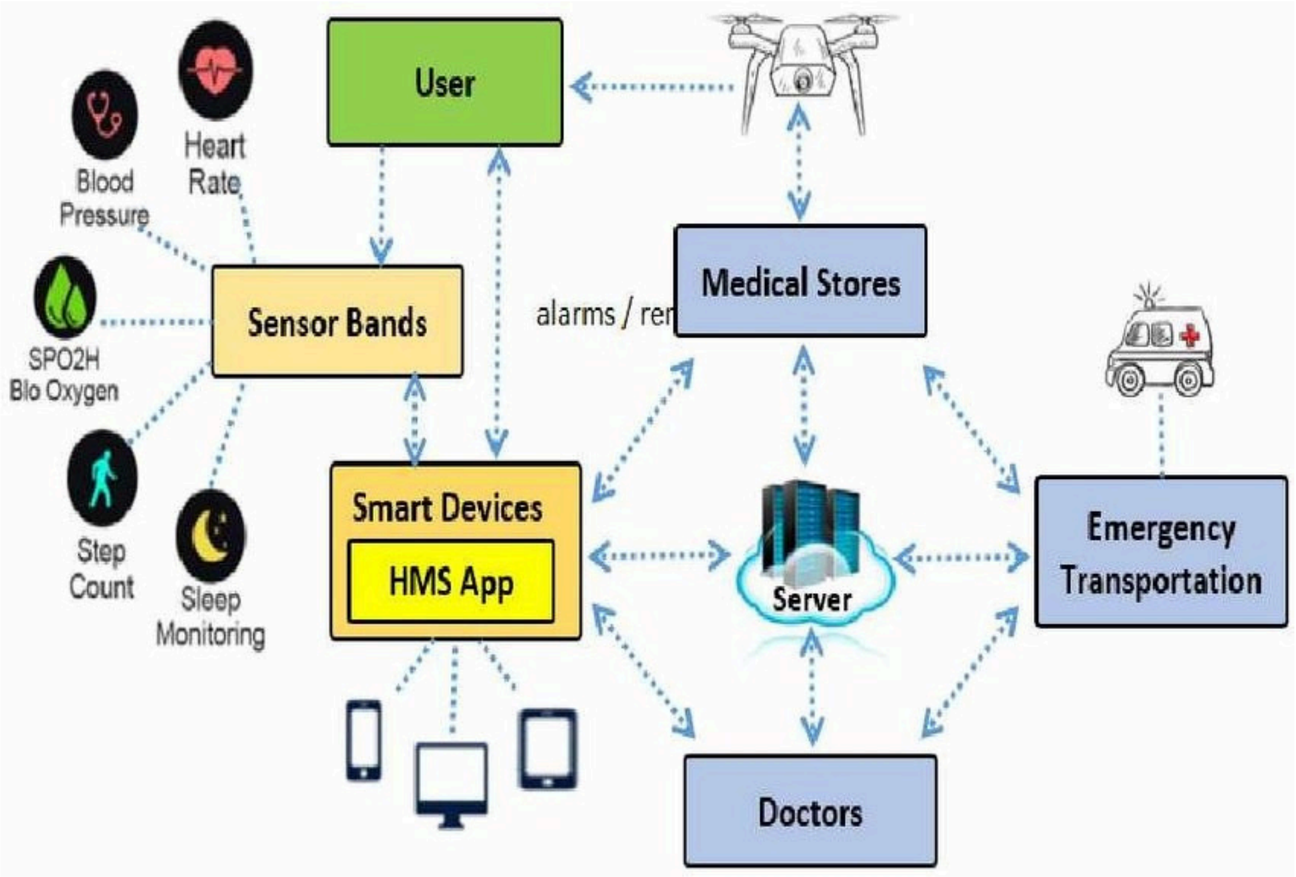
Applications

Smart Cities: Manage traffic, waste, street lighting, and environmental monitoring to improve urban living.



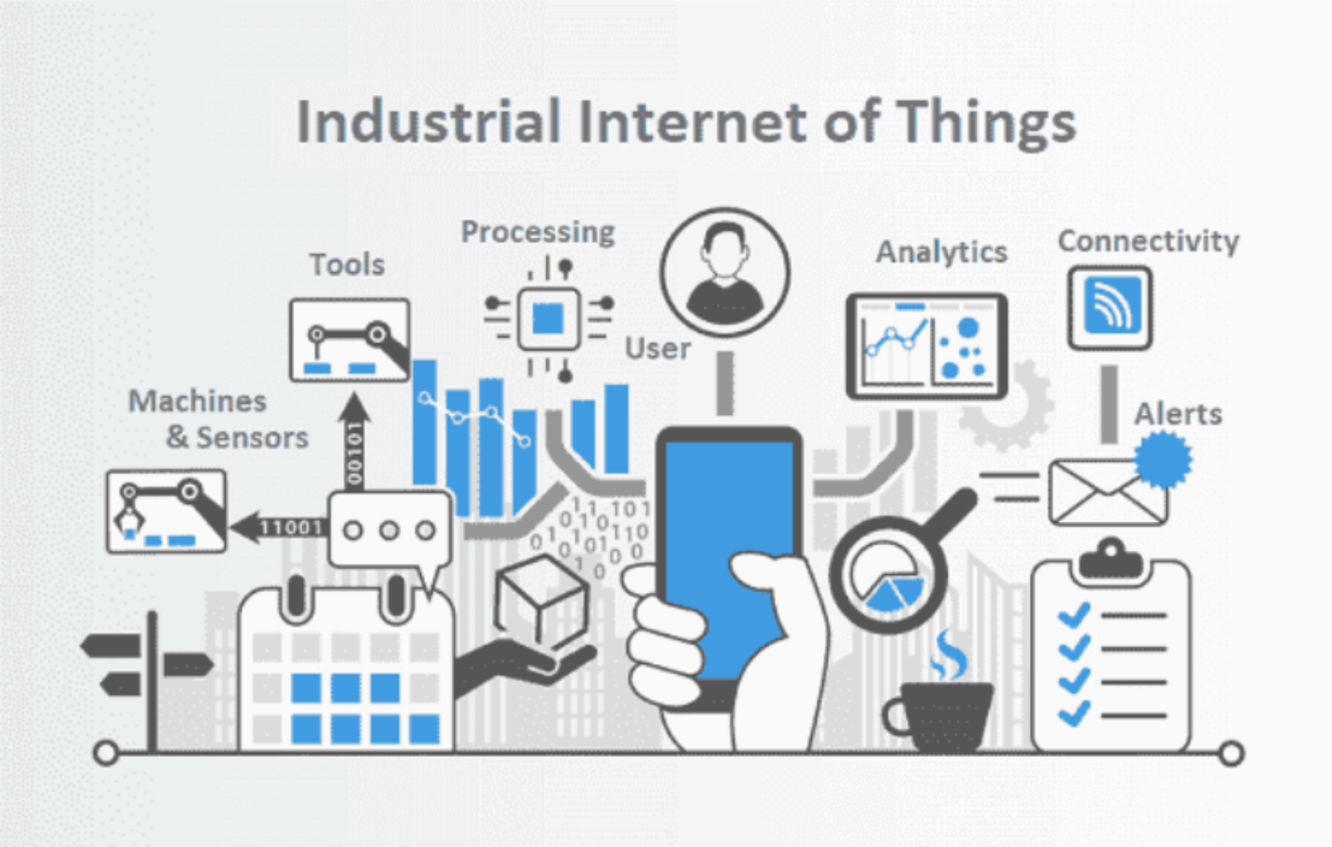
Applications

Healthcare and Wearables: Track health metrics, monitor patients remotely, and provide personalized care.



Applications

Industrial IoT (IIoT): Optimize factory operations with predictive maintenance, automation, and machine-to-machine communication.

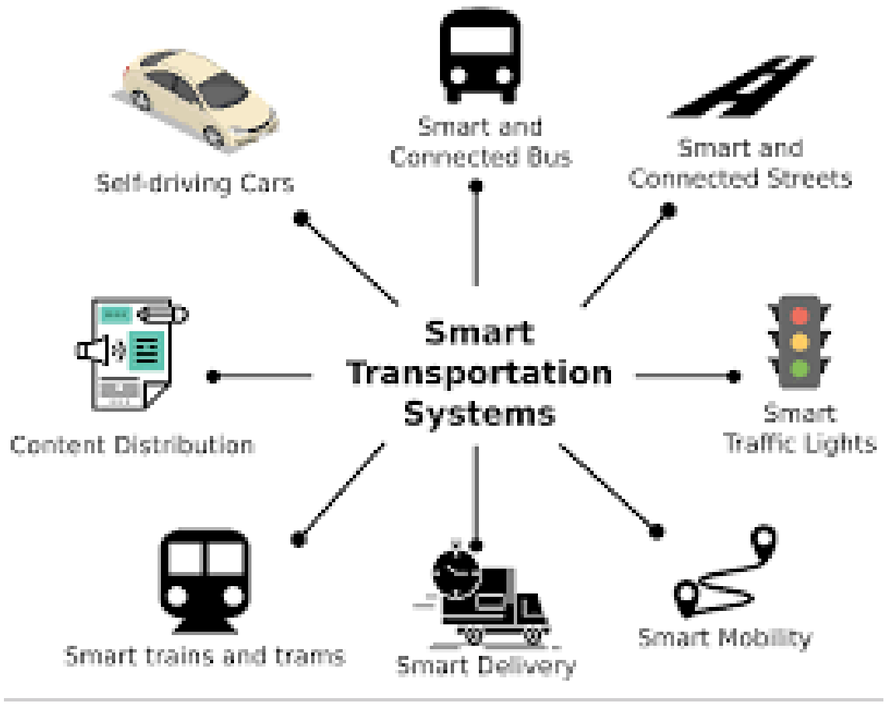


Applications

Agriculture: Monitor soil, crops, and irrigation systems to increase yield and reduce resource waste.

Applications

Transportation and Logistics: Track vehicles, manage fleets, monitor cargo, and optimize delivery routes.



Resources

GFG Blog

Acknowledgement

Dr. A. B. M. Alim Al Islam

Professor

CSE, BUET

Tech Enthusiasts: Everything in my house is wired to the Internet of Things! I control it all from my smartphone! My smart-house is bluetooth enabled and I can give it voice commands via alexa! I love the future!

Programmers / Engineers: The most recent piece of technology I own is a printer from 2004 and I keep a loaded gun ready to shoot it if it ever makes an unexpected noise.

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